

Comments on “*Deep Fréchet Regression*” by Su I Iao, Yidong Zhou & Hans-Georg Müller

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Abstract

We provide a written commentary of the article *Deep Fréchet Regression* by [Iao et al. \(2025\)](#), which was presented at the JASA Theory and Methods Invited Session at the 2026 Joint Statistical Meetings (JSM).

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1 Introduction

We congratulate the authors on their impressive *Deep Fréchet Regression* (DFR) article, which further enriches a general and substantive program on modeling complex object responses using Fréchet methods, emanating from the seminal work of [Petersen and Müller \(2019\)](#). In recent years, there has been an increase in the volume, complexity and diversity of data objects being collected. Many of these modern object types have non-Euclidean, complex high-dimensional structure, e.g., networks ([Desai et al., 2025](#); [Zhou and Müller, 2022](#)), probability distributions ([Kneip and Utikal, 2001](#); [Petersen and Müller, 2016](#); [Yang et al., 2020](#); [Maier et al., 2025](#)), shape data ([Wu et al., 2024](#)), and brain cortical surfaces ([Chung et al., 2022](#)). The development of principled statistical approaches that leverage the objects’ inherent structure and obey their geometric constraints is a very important endeavor. Moreover, we are glad to see the authors combine modern deep-learning methods within classical statistical approaches, thereby developing a “hybrid” framework that benefits from the strengths of both strategies.

The authors have developed an impressively broad and flexible approach for regressing non-Euclidean object data on multivariate Euclidean predictors. In theory, the only requirement is that the objects reside in a metric space, i.e., that a suitable distance is defined for them. This distance is employed in a first-stage manifold learning of the object-valued responses, to embed them in a lower-dimensional latent space that aims to preserve their geodesic distances. The authors then employ non-parametric regression using modern deep learning to regress the latent embeddings on the multivariate covariates. Again, this non-parametric regression step is very general, and allows them to flexibly capture nonlinearities and interaction effects without imposing parametric assumptions. The fitted values from the latent non-parametric regression are subsequently used as low-dimensional predictors in a local Fréchet regression step, thereby leveraging the structure learned in the

latent non-parametric regression with the Fréchetmapping using the appropriate distance metric and ensuring that the objects’ geometric constraints are preserved in all predictions. The generality of the proposed methodology means that it can be applied, as the authors demonstrate in their examples, to connectivity matrices and to univariate probability distributions alike without any modifications, and without making any restrictive assumptions on the form of covariate effects. To our best understanding, the method is also readily applicable to, for example, spherical and compositional data.

The theoretical results provide a solid grounding for the framework and is one the works’s strengths. The authors establish convergence rates for DFR that account for the error in both the modeling of the latent embeddings and in the local Fréchetregression step. The data application, where DFR is used to model the effect of meteorological variables, days of the week, and aggregated service-related covariates on New York City taxi networks is another strength of the work. This interesting example highlights the practical applicability of the framework to modern non-Euclidean data.

The remainder of our commentary relates to issues regarding the practical implementation of the DFR framework, that we believe are worthy of further discussion.

2 Practical Considerations for Applying the DFR Framework

2.1 Selecting the Intrinsic Manifold Dimension

At present, the authors employ the ISOMAP technique as the default dimension reduction method in both their manuscript and code¹, though their network data application employs Multidimensional Scaling (MDS) to represent the graph Laplacian objects. The dimension

¹<https://github.com/SUIIAO/Deep-Frechet-Regression>

of the latent embedding space, which the authors denote by r , is fixed at $r = 2$ for all three examples—the distributional simulation study, the traffic networks example, and the mortality distributions application in the supplementary material—and the current software implementation only permits $r \leq 2$. For the simulation study, $r = 2$ is certainly a suitable choice because the simulated data, by construction, reside on a two-dimensional manifold, i.e., within the space of Gaussian distributions. Still, this true intrinsic dimension would not be known *a priori* in a real data analysis setting. While the $r = 2$ assumption seems reasonable in the simulations and case studies presented in the main text and appendices, more generally it is not clear whether it will always be a sensible default, nor whether the methodology is sensitive to mis-specification of this dimension.

This dimension reduction step plays a crucial role in the proposed methodology. The low-dimensional latent Euclidean embeddings are treated as surrogates for the original objects, allowing them to be regressed non-parametrically on Euclidean predictors using deep learning. The fitted values from this regression are then used as predictors within the local Fréchetregression. Motivated in this way, it seems reasonable that an effective dimension reduction method should have two main properties. Firstly, the latent embeddings should preserve all (or most) of the structure in the original object space, such that the relationship with the predictors can be faithfully captured in the non-parametric regression. Second, since the fitted embeddings are used as predictors in the Fréchetregression step, we want them to be as low-dimensional (*compact*) as possible, to avoid the curse of dimensionality in local regression, which is captured by the derived convergence rate for multivariable local Fréchetregression in Proposition 1 of the manuscript. In our own work (Compact near-Lossless Latent Representations (CLaRe); [Zohner et al., 2025](#)), we call these two properties **(near-)losslessness** and **compactness**. In practice, they can be regulated through both the choice of dimension reduction method and the number of latent features retained, i.e., r , the dimension of the lower-dimensional embedding space.

Currently, due to the general metric structure, DFR employs non-invertible manifold learning techniques for dimension reduction that require only a pairwise distance matrix between observations (e.g., ISOMAP, MDS). In contrast, our CLaRe framework focuses on dimension reduction methods with explicit encoding-decoding architectures,. We use the generalization error of decoded objects to measure of how well their structure is being captured. For non-invertible manifold learning techniques, an analogue might be how well the objects’ geodesic distances are preserved by the lower-dimensional embeddings. In CLaRe we argue that to confidently use the embeddings as surrogates for the individual objects in downstream analysis, we should aim to control (out-of-sample) individual observation-level reconstruction error, rather than aggregated measures. In the manifold learning context, one might examine the geodesic distance reconstructions for each object individually, rather than aggregate them over the whole dataset, although we must assume some further structure to translate this into lossless reconstruction of the original data objects, something to which we will return in the next section.

To translate our point on manifold selection into a practical example, we have built on the authors’ simulation scenario from the main paper and supplementary material. We simulated a dataset of size $n = 500$ from their original Gaussian data setting and from their perturbed Gaussian setting, using these as references. We then simulated a dataset in the same way from a *skew-normal* distribution, parameterized such that the random means and variances of each distribution are drawn from the same common population distributions as in the authors’ examples, but these new random distributions additionally depend on a skew parameter α (Azzalini, 2013). The conditional expectation of the α parameter was generated as a non-linear function of the generated covariates, such that it had a non-linear relationship with the expected mean and variance. We also interacted it with the binary covariate X_9 , so that it could reflect a realistic group effect.

The top row of Figure 1 displays the three datasets, represented in terms of their

discretized empirical quantile functions. On the surface, all three sets of discretized empirical quantile functions appear reasonably similar and it is not immediately obvious that $r > 2$ might be required for the third dataset, but not for the first and second. We then linearly projected the discretized quantile functions onto three dimensions that approximately represent the mean, variance, and skewness of the distributions, respectively. This three-dimensional representation is shown in the second row of Figure 1, colored according to the feature X_9 . From these plots, the difference in dimensionality is more evident—for the skew-normal example, there is additional structured variation in the skewness (vertical) coordinate, which exhibits a between-group/covariate dependence. The third and fourth rows show some diagnostics that could be used to identify a suitable ISOMAP manifold dimension, r . The third row displays the first ten eigenvalues from the MDS step in ISOMAP; we interpret these as an aggregated measure of how well an r -dimensional ISOMAP representation preserves the geodesic distances in the dataset. More in line with our CLaRe philosophy, in the bottom row (left) we present a measure of how well each of the 500 observation’s geodesic distances with the other objects is preserved, for $r = 1, \dots, 4$. This plot makes it more clear that 3 dimensions should be retained with ISOMAP to preserve the structure in the original space, since the distance reconstruction errors almost vanish when moving from a two to three-dimensional representation. Finally, on the right of the bottom panel, we plot the third ISOMAP dimension against the true expected α parameter, very clearly this dimension is required for capturing the between group difference. If the default $r = 2$ was used, this effect would not be retained in the embeddings and therefore missed by the non-parametric regression. This short example demonstrates that careful selection of the embedding dimension r is an important step of the DFR methodology and should be given further consideration.

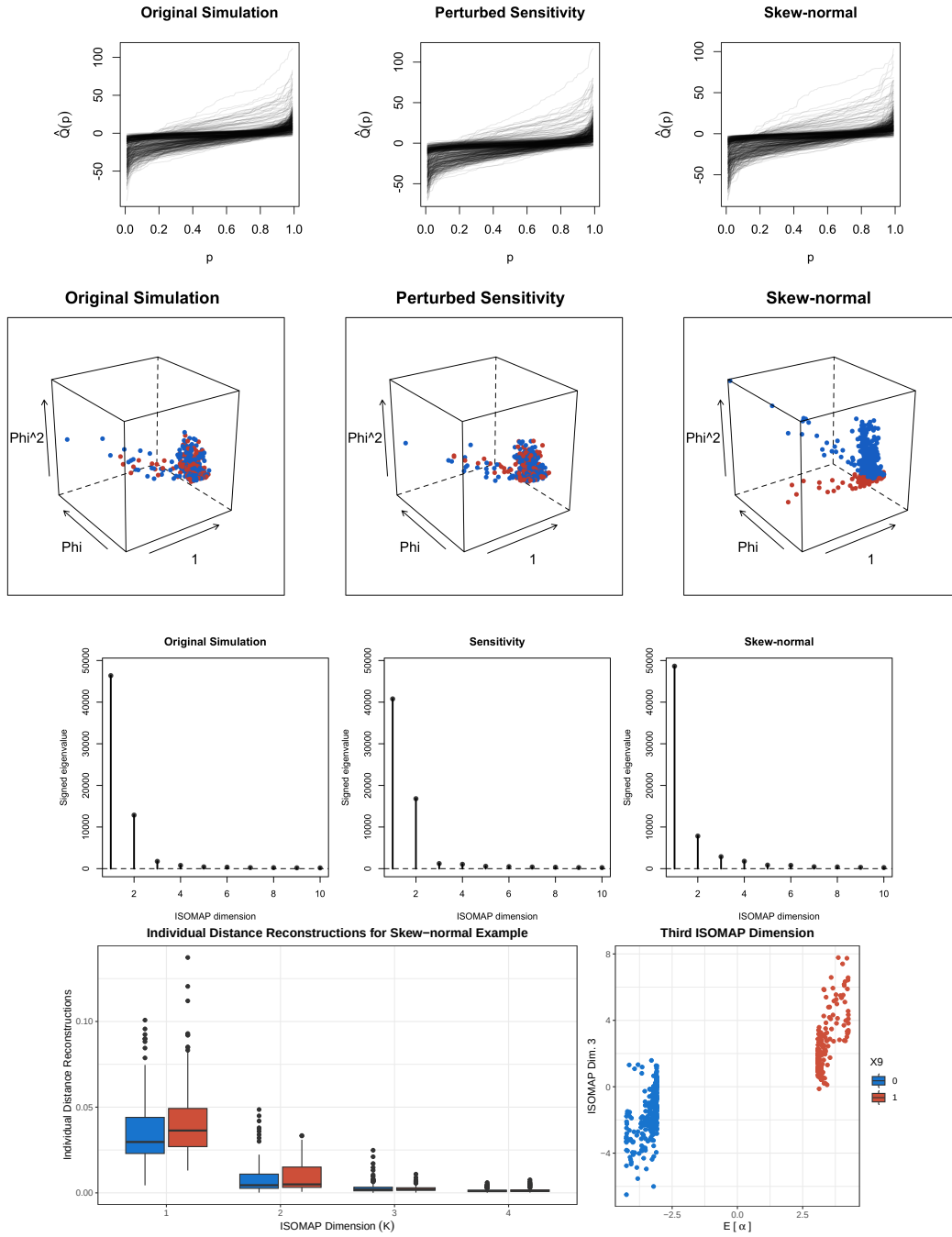


Figure 1: A selection of figures from our demonstration on selecting the manifold dimension selection, based on data simulated from the authors’ main simulation setting, their perturbed sensitivity analysis and our own modification including skew-normal data. **First row:** The empirical quantile functions from the authors’ main simulation, their sensitivity analysis and our skew normal (left, center and right, respectively). **Second row:** The linear projection of the empirical quantile functions into three-dimensional space that approximately spans the skew-normal family for each of the examples. Here points are colored according to their value of the binary covariate X_9 . **Third row:** A plot of the eigenvalues obtained from the MDS step in ISOMAP for each of the models. For the skew-normal (left column), the third eigenvalue is

2.2 Does Generality Preclude Encoder-Decoder Architectures?

Returning to the analogy of the local Fréchetregression as a decoder, one of the attractive features of DFR is that it only requires the response objects to live in a space equipped with a metric. In practice, however, computation of the local Fréchetregression model involves a search over the space of objects, which implies the need for additional structure. Structure, in this context, refers to how the objects are represented numerically in a vector (i.e., “vectorized”) in order to perform computations. Object-specific examples include the discretized quantile function representation of distributions and the positive semidefinite matrix parametrization of covariance objects.

Once a vectorization structure for the objects has been assumed, we are back to working within a constrained subset of Euclidean space, where explicit encoder-decoder architectures could be applied. On a case-by-case basis, we can often find a transformation that maps the vectorized representations to unconstrained Euclidean space, e.g., the log-quantile density transformation for quantile functions ([Petersen and Müller, 2016](#)) and the multivariate generalization of Fisher’s transformation for correlation matrices ([Desai et al., 2025](#)). Such transformations can be used to compose bespoke encoder-decoder architectures for different object types, at the expense of the full generality that DFR provides.

The advantage of employing a transformation that decodes objects is that it allows us to eliminate, reduce, or at least diagnose approximation error in the latent space due to object reconstruction, isolating error purely from the latent non-parametric model. At present, DRF has approximation error from both sources, and although it is accounted for in the authors’ theory, it can be challenging to assess in practice, as highlighted in the skew-normal example above. In our own framework that is applied to images ([Gunning et al., 2026](#)), we employ a near-lossless encoder-decoder architecture and a parametric latent regression model. Doing so constrains approximation error to the familiar Euclidean setting, where

classical assessments and diagnostics can be employed. It is interesting to contrast this with subsequent developments to the Fréchetregression framework, where the dual error source is eliminated by applying deep learning directly to learn the regression weights (CITE END TO END). Regardless, it is worth considering the merits of, on a case-by-case basis, harnessing the additional vectorization structure to employ explicit encoder-decoder architectures in the DRF framework.

2.3 Inference, Stability and Scalability

We raise a few further points in relation to the practical application of the DFR framework, while noting that these are already being addressed within the broader Fréchet regression program. The first point regards *inference*. The authors’ real data application to New York City (NYC) taxi trip networks demonstrates that the method has *descriptive*, as well as *predictive*, potential. The display and description of the precipitation effect on predicted traffic networks, varying across different days of the week, is presented in an interesting and interpretable way. However, it is not clear whether this effect, shown on the heatmaps, can be deemed “statistically significant” or whether uncertainty regions for it can be constructed. To this end, we are happy to see inferential approaches for Fréchetregression being developed by [Song et al. \(2026\)](#).

Secondly, the flexibility of the deep neural network regression is one of the paper’s strengths. However, when a parsimonious, e.g., parametric or semiparametric, specification of covariate effects is sufficient, this flexibility may introduce unnecessary complexity or instability. Again, related developments, such as deep single-index Fréchet regression ([Cui et al., 2026](#)), offer more structured parameterizations that may be beneficial in such cases. Guidance on choosing between flexible and structured parameterizations, especially with respect to efficiency, computational feasibility, and covariate dimension, would be a welcome addition.

3 Conclusion

To summarize, the authors are to be congratulated on developing a very general strategy for regressing complex object data on Euclidean predictors. The main strength of the methodology is its generality and flexibility, as it imposes minimal assumptions on the object-valued responses and the effects of covariates. The method is supported by strong theoretical results and exemplified using an interesting application to traffic network data. We believe that the practical applicability of the method could be improved by guidance on selecting the dimension reduction architecture and the intrinsic manifold dimension, for which we have shared some perspectives from our own work employing conventional encoder-decoder architectures. We also envision that some consideration of inference, scalability and stability, within the context of concurrent developments in the general Fréchet regression program, will further enhance its impact.

References

Azzalini, A. (2013), *The Skew-Normal and Related Families*, Institute of Mathematical Statistics Monographs, Cambridge University Press, Cambridge.

URL: <https://www.cambridge.org/core/books/skewnormal-and-related-families/4B9A3CD9D58EB6022FC1D21E1E08A10E>

Chung, M. K., Hanson, J. L. and Pollak, S. D. (2022), ‘Statistical analysis on brain surfaces’, *arXiv preprint arXiv:2203.06665*.

URL: <https://arxiv.org/abs/2203.06665>

Cui, M., Zhou, Y., Iao, S. I. and Müller, H.-G. (2026), ‘Deep Single-Index Fréchet Regression’. *arXiv:2606.06957 [stat.ML]*.

URL: <http://arxiv.org/abs/2606.06957>

Desai, N., Baladandayuthapani, V., Shinohara, R. T. and Morris, J. S. (2025), ‘Connectivity Regression’, *Biostatistics* **26**(1), kxaf002.

URL: <https://doi.org/10.1093/biostatistics/kxaf002>

Gunning, E., Hooker, G. and Morris, J. S. (2026), Statistical generative regression models for structured object data. Manuscript in preparation.

Iao, S. I., Zhou, Y. and Müller, H.-G. (2025), ‘Deep Fréchet Regression’, *Journal of the American Statistical Association* **120**(551), 1437–1448.

URL: <https://doi.org/10.1080/01621459.2025.2507982>

Kneip, A. and Utikal, K. J. (2001), ‘Inference for Density Families Using Functional Principal Component Analysis: Rejoinder’, *Journal of the American Statistical Association* **96**(454), 540–542.

URL: <https://www.jstor.org/stable/2670295>

Maier, E.-M., Stöcker, A., Fitzenberger, B. and Greven, S. (2025), ‘Additive density-on-scalar regression in Bayes Hilbert spaces with an application to gender economics’, *The Annals of Applied Statistics* **19**(1), 680–700.

URL: <https://projecteuclid.org/journals/annals-of-applied-statistics/volume-19/issue-1/Additive-density-on-scalar-regression-in-Bayes-Hilbert-spaces-with/10.1214/24-AOAS1979.full>

Petersen, A. and Müller, H.-G. (2016), ‘Functional data analysis for density functions by transformation to a Hilbert space’, *The Annals of Statistics* **44**(1), 183–218.

URL: <https://projecteuclid.org/journals/annals-of-statistics/volume-44/issue-1/Functional-data-analysis-for-density-functions-by-transformation-to-a/10.1214/15-AOS1363.full>

Petersen, A. and Müller, H.-G. (2019), ‘Fréchet regression for random objects with Euclidean predictors’, *The Annals of Statistics* **47**(2).

URL: <https://doi.org/10.1214/17-AOS1624>

Song, W., Dubey, P., Müller, H.-G. and Petersen, A. (2026), ‘Inference for Fréchet Regression’. arXiv:2605.19519 [stat.ME].

URL: <http://arxiv.org/abs/2605.19519>

Wu, Y., Huang, C. and Srivastava, A. (2024), ‘Shape-based functional data analysis’, *TEST* **33**(1), 1–47.

URL: <https://doi.org/10.1007/s11749-023-00876-9>

Yang, H., Baladandayuthapani, V., Rao, A. U. and Morris, J. S. (2020), ‘Quantile Function on Scalar Regression Analysis for Distributional Data’, *Journal of the American Statistical Association* **115**(529), 90–106.

URL: <https://doi.org/10.1080/01621459.2019.1609969>

Zhou, Y. and Müller, H.-G. (2022), ‘Network Regression with Graph Laplacians’, *Journal of Machine Learning Research* **23**(320), 1–41.

URL: <http://jmlr.org/papers/v23/22-0681.html>

Zohner, E., Gunning, E., Hooker, G. and Morris, J. (2025), ‘CLaRe: Compact near-lossless Latent Representations of High-Dimensional Object Data’. arXiv:2502.07084 [stat].

URL: <http://arxiv.org/abs/2502.07084>